

A Project to Put New Quads Downstream of the Møller setup

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There are several requirements to the optics settings in Hall A:

1. A good beam spot on the main target
2. A small beam spot in "X" projection in the Compton polarimeter chicane
3. Settings in 3 "Møller" quads optimized for Møller polarimeter measurements

With the present beam line configuration there is no way to satisfy all the requirements at the same time. Changing the magnets to/from the Møller settings takes on average 20min. In order to provide a common setting satisfying all the requirements the accelerator division suggested to install 2 new quads downstream of the Møller dipole.

There is a potential drawback of such a change in the beam line configuration. Such quads may increase the background for the Møller detector. At the moment the beam pipe downstream of the Møller dipole is 4" thick. It turns to a 1.5" pipe about 200cm downstream of the end of a Møller detector box, of about 1000cm from the Møller target. Multiple scattering in the Møller target smears off the beam considerably. For a 0.8GeV beam the deflection is about $RMS=1cm$ at 1000cm from the Møller target.

A quad with a 1" beam hole placed above the Møller detector may create a considerable background since 10-40% of the beam passed